

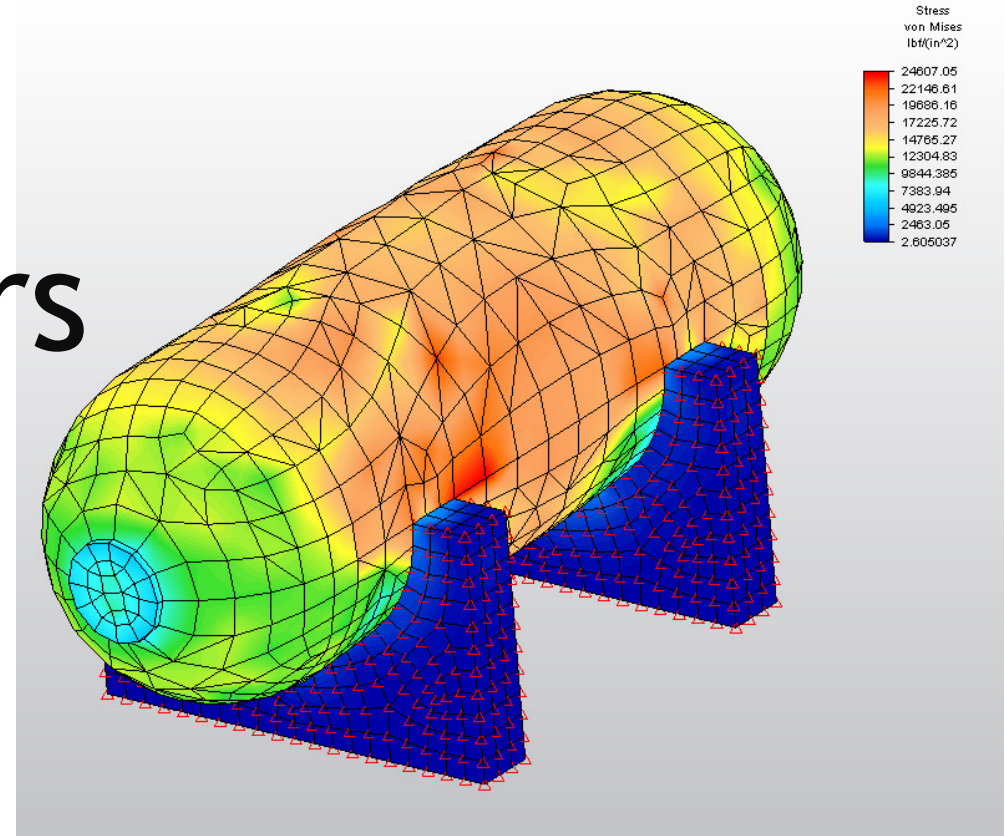


Strength of Materials II



Chapter Three: Thin & Thick Cylinders

MEng3111



Thin Cylinder

- A vessel either cylindrical or spherical in shape used to carry a fluid (gas or liquid) under pressure is known as a pressure vessel.
- A Cylinder used to store pressurized fluid whose wall thickness is less than or equal to $1/15$ of its internal radius is known as a thin cylinder.
- If the wall thickness is more than one $1/15$ of the internal radius it is considered as a thick cylinder.

Thin Cylinders

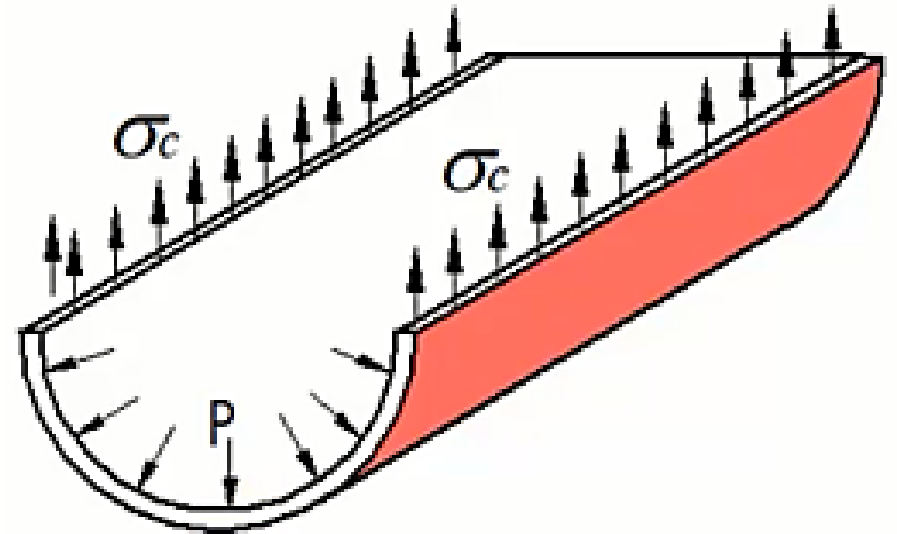
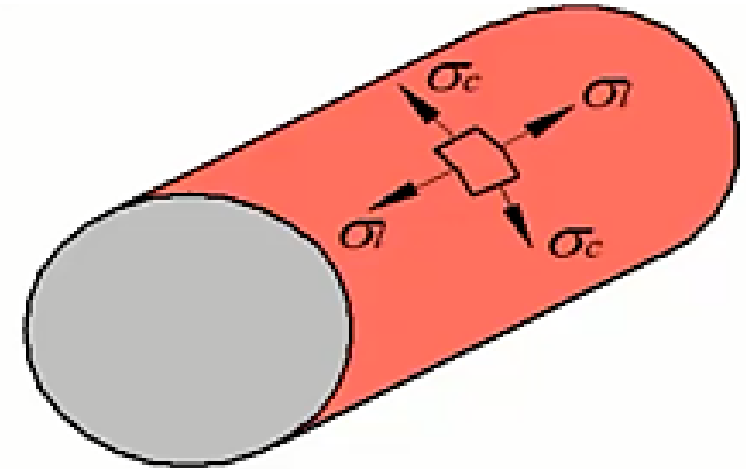
The projected area over which fluid pressure acts= pd

- Bursting force= pdl
- Resisting area= $2tl$
- Hoop stress or circumferential stress,

$$\begin{aligned}\sigma_c &= \frac{\text{Bursting force}}{\text{Resisting area}} \\ &= \frac{pdl}{2tl} \\ &= \frac{pd}{2t}\end{aligned}$$

Hoop stress, $\sigma_c = \frac{pd}{2t}$

----- (1)



Thin Cylinders

Axial force due to fluid pressure,

$$F_l = P \times \frac{\pi}{4} d^2$$

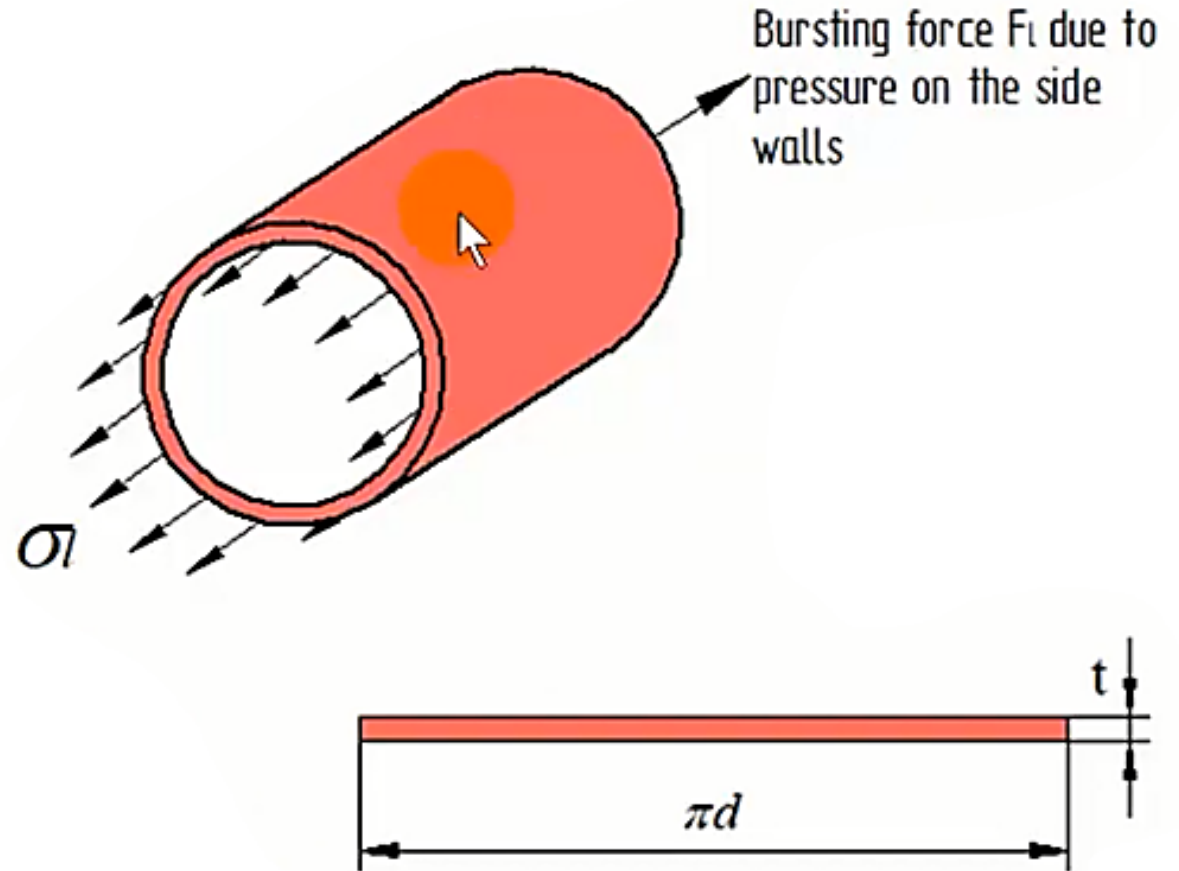
Resisting area $A = \pi dt$

Longitudinal stress, $\sigma_l = \frac{F_l}{A}$

$$= \frac{p \times \frac{\pi}{4} d^2}{\pi dt} = \frac{pd}{4t}$$

Longitudinal stress, $\sigma_l = \frac{pd}{4t}$

 ----- (2)

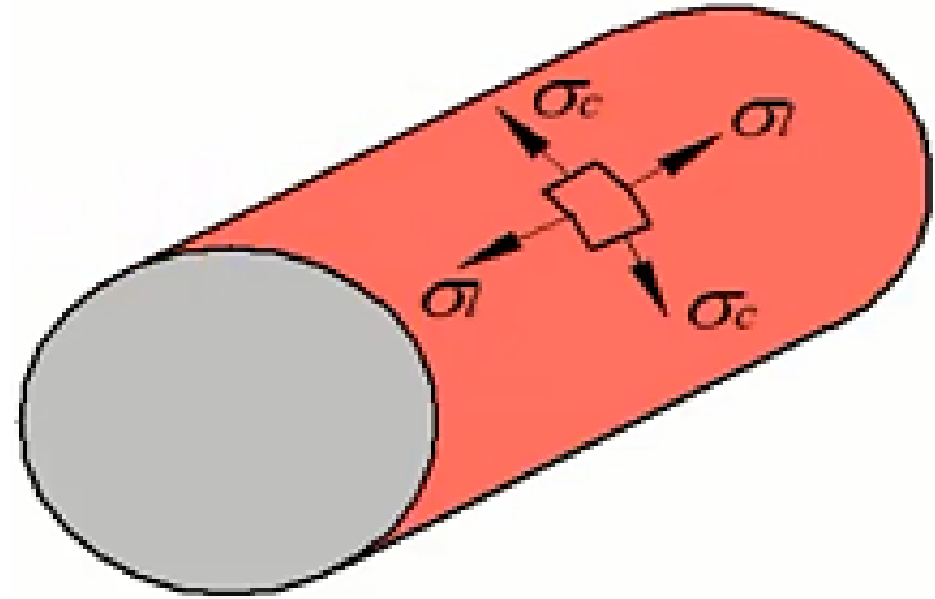


Thin Cylinders

Circumferential or Hoop stress, $\sigma_c = \frac{pd}{2t}$

Longitudinal stress, $\sigma_l = \frac{pd}{4t}$

Hence, $\sigma_l = \frac{\sigma_c}{2}$



Also we can show that maximum shear stress, $\tau_{\max} = \frac{\sigma_c - \sigma_l}{2}$

$$\tau_{\max} = \frac{pd}{8t} = \frac{\sigma_c}{4} = \frac{\sigma_l}{2}$$

Example 1

A thin cylinder 300mm internal diameter and 10mm thick wall is subjected to internal fluid pressure 6Mpa. Find the longitudinal, Hoop stress and maximum shear stress induced in the cylinder.

- Data Given:
- $P=6\text{MPa}$
- $d=300\text{mm}$
- $t=10\text{mm}$

Example 1

$$\begin{aligned}\text{Longitudinal stress, } \sigma_l &= \frac{pd}{4t} \\ &= \frac{6 \times 300}{4 \times 10} \\ &= 45 \text{ MPa}\end{aligned}$$

$$\begin{aligned}\text{Circumferential stress, } \sigma_c &= \frac{pd}{2t} \\ &= \frac{6 \times 300}{2 \times 10} \\ &= 90 \text{ MPa}\end{aligned}$$

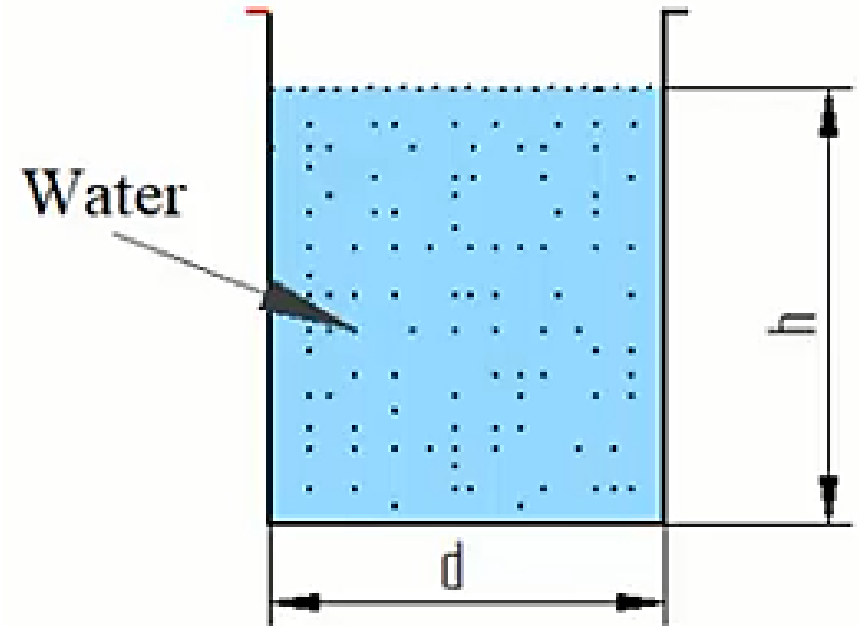
Example 1

$$\begin{aligned}\text{Maximum shear stress, } \tau_{\max} &= \frac{Pd}{8t} \\ &= \frac{6 \times 300}{8 \times 10} \\ &= 22.5 \text{ MPa}\end{aligned}$$

Example 2

A city water reservoir with 0.4m diameter supplies water at a height of 80m. Determine the thickness of the pipe taking an allowable stress for the material as 25MPa. Specific weight of water is 9.81kN/m³.

Water pressure,
$$p = \gamma h = 9810 \times 80 = 784800 \text{ N/m}^2$$
$$= 0.7848 \text{ N/mm}^2$$



Example 2

- $d=0.4\text{m}=400\text{mm}$
- Specific weight, $\gamma = 9.81\text{KN/m}^3 = 9810\text{N/m}^3$
- Allowable stress = 25Mpa
- (This stress has to be taken as the Hoop stress, since its value is always higher in a thin cylinder.)

$$\text{Circumferential stress, } \sigma_c = \frac{pd}{2t}$$

$$25 = \frac{0.7848 \times 400}{2 \times t}$$

$$t = 6.2784 \text{ mm}$$

Example 3

A water pipe with 600mm diameter supplies water at a head of 62m. Determine the thickness of the pipe required when a material having yield stress 200MPa is used. Given, Specific weight of water is 9.81 kN/m³ and factor of safety 4.

Example 3

$$d=600\text{mm}$$

$$\text{Specific weight, } \gamma = 9.81\text{KN} / \text{m}^3 = 9810\text{N} / \text{m}^3$$

$$\text{Yield stress} = 200\text{MPa}$$

$$\text{Allowable stress} = \frac{\text{Yield stress}}{\text{Factor of safety}} = \frac{200}{4} = 50\text{MPa}$$

(This stress has to be taken as the Hoop stress, since its value is always higher in a thin cylinder.)

$$\begin{aligned}\text{Water pressure, } p &= \gamma h = 9810 \times 62 = 608220 \text{ N/m}^2 \\ &= 0.60822 \text{ N/mm}^2\end{aligned}$$

Example 3

Circumferential stress, $\sigma_c = \frac{pd}{2t}$

$$50 = \frac{0.60822 \times 600}{2 \times t}$$
$$t = 3.649 \text{ mm}$$

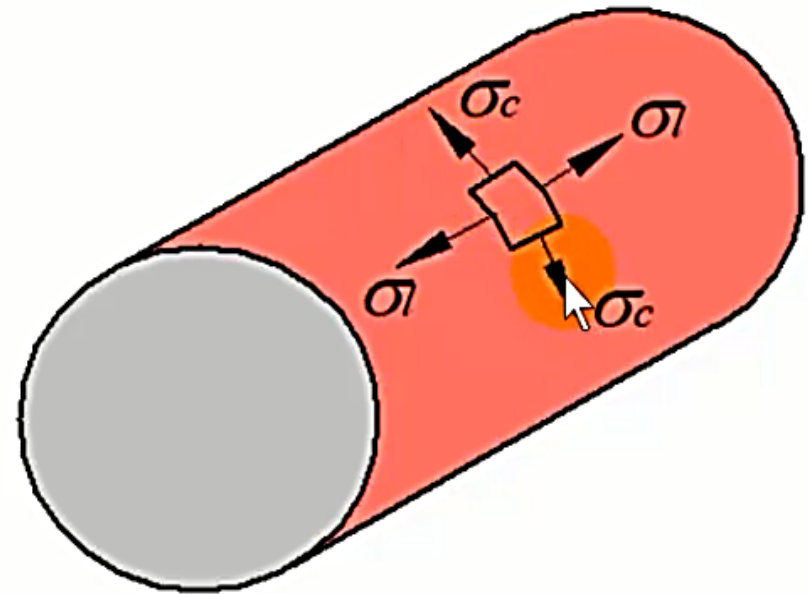
Longitudinal, circumferential & volumetric strain

Change in dimensions of a thin cylinder

The stresses induced increases the circumference and length of the cylinder.

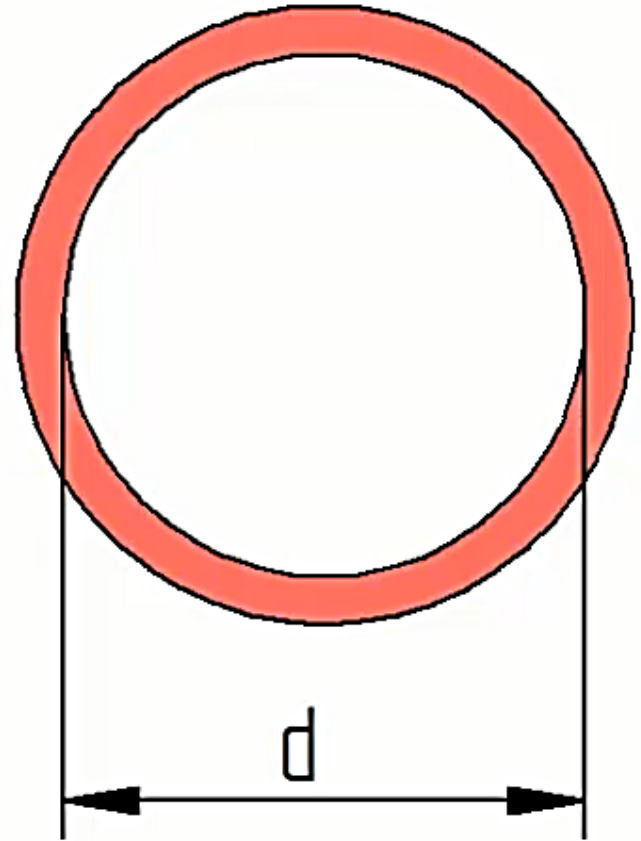
$$\text{Circumferential strain} = \frac{\text{Change in circumference}}{\text{Original circumference}}$$

The circumferential change is caused due to the changes in diameter.

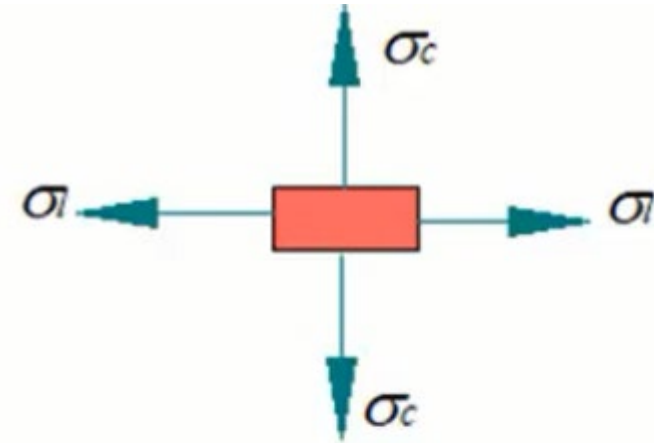


Let the initial diameter of the cylinder is 'd' and the final diameter is $d+\delta d$.

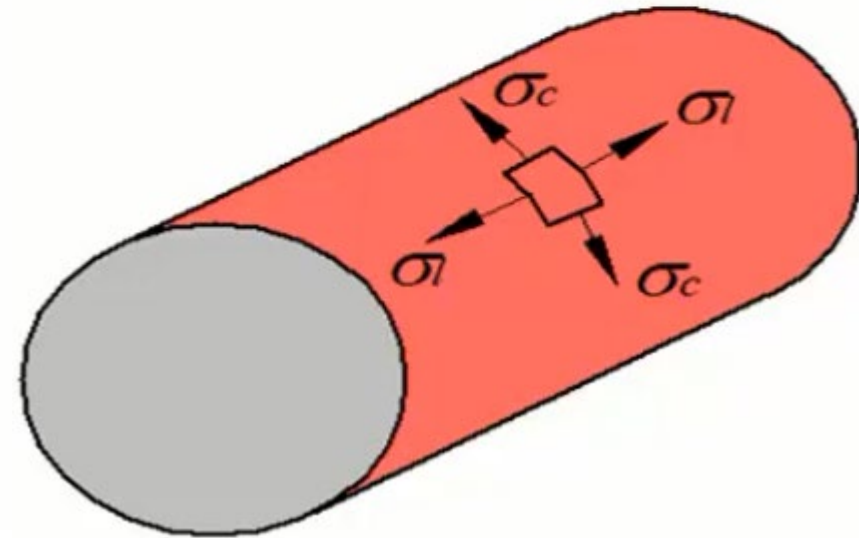
$$\begin{aligned}\text{Circumferential strain} &= \frac{\pi(d+\delta d) - \pi d}{\pi d} \\ &= \frac{\delta d}{d} \\ &= \text{strain in diameter}\end{aligned}$$



$$\begin{aligned}
 \text{Longitudinal strain, } e_l &= \frac{\sigma_l}{E} - \nu \frac{\sigma_c}{E} \\
 &= \frac{pd}{4tE} - \nu \frac{pd}{2tE} \\
 &= \frac{pd}{4tE} (1 - 2\nu)
 \end{aligned}$$



$$\begin{aligned}
 \text{Circumferential strain, } e_c = e_d &= \frac{\sigma_c}{E} - \nu \frac{\sigma_l}{E} \\
 &= \frac{pd}{2tE} - \nu \frac{pd}{4tE} \\
 &= \frac{pd}{4tE} (2 - \nu)
 \end{aligned}$$



Volumetric strain, $e_v = e_l + 2e_d$

$$\begin{aligned} &= \left(\frac{\sigma_l}{E} - \nu \frac{\sigma_c}{E} \right) + 2 \left(\frac{\sigma_c}{E} - \nu \frac{\sigma_l}{E} \right) \\ &= \frac{pd}{4tE} - \nu \frac{pd}{2tE} + 2 \left(\frac{pd}{2tE} - \nu \frac{pd}{4tE} \right) \\ &= \frac{pd}{4tE} + \frac{pd}{tE} - \nu \frac{pd}{tE} \\ &= \frac{5pd}{4tE} - \nu \frac{pd}{tE} \\ &= \frac{pd}{tE} \left(\frac{5}{4} - \nu \right) = \frac{pd}{4tE} (5 - 4\nu) \end{aligned}$$

$$\Delta V = e_v \times V = \frac{pd}{4tE} (5 - 4\nu) \times \frac{\pi}{4} d^2 l = \frac{\pi p d^3 l}{16tE} (5 - 4\nu)$$

Example 4

A thin cylinder 300mm internal diameter, 1m long and 12mm thick is subjected to internal fluid pressure of 5MPa. Find the change in length, diameter and volume of cylinder. Take $\nu=0.25$ and $E=200\text{GPa}$.

Example 4

$$\begin{aligned}\text{Circumferential stress, } \sigma_c &= \frac{pd}{2t} \\ &= \frac{5 \times 300}{2 \times 12} \\ &= 62.5 \text{ MPa}\end{aligned}$$

$$\begin{aligned}\text{Longitudinal stress, } \sigma_l &= \frac{pd}{4t} \\ &= \frac{5 \times 300}{4 \times 12} \\ &= 31.25 \text{ MPa}\end{aligned}$$

Example 4

$$\begin{aligned}\text{Diameter strain, } e_d &= \frac{\sigma_c}{E} - \nu \frac{\sigma_l}{E} \\ &= \frac{1}{E} (\sigma_c - \nu \sigma_l) \\ &= \frac{1}{200 \times 10^3} (62.5 - 0.25 \times 31.25) \\ &= 0.00027344\end{aligned}$$

$$\begin{aligned}\text{Change in length, } \delta l &= e_l \times l \\ &= 7.813 \times 10^{-5} \times 1000 \\ &= 0.07813 \text{ mm}\end{aligned}$$

Example 4

$$\begin{aligned}\text{Longitudinal strain, } e_l &= \frac{\sigma_l}{E} - \nu \frac{\sigma_c}{E} \\ &= \frac{1}{E} (\sigma_l - \nu \sigma_c) \\ &= \frac{1}{200 \times 10^3} (31.25 - 0.25 \times 62.5) \\ &= 0.00007813\end{aligned}$$

$$\begin{aligned}\text{Change in diameter } \delta d &= e_d \times d \\ &= 2.7344 \times 10^{-4} \times 300 \\ &= 0.082 \text{ mm}\end{aligned}$$

Example 4

Volumetric strain, $e_v = e_l + 2e_d$

$$\begin{aligned} &= 7.813 \times 10^{-5} + 2 \times 2.7344 \times 10^{-4} \\ &= 0.00062501 \end{aligned}$$

Change in volume, $\delta v = e_v \times v$

$$\begin{aligned} &= e_v \times \frac{\pi}{4} d^2 l \\ &= 0.00062501 \times \frac{\pi}{4} 300^2 \times 1000 \\ &= 44179.354 \text{ mm}^3 \end{aligned}$$

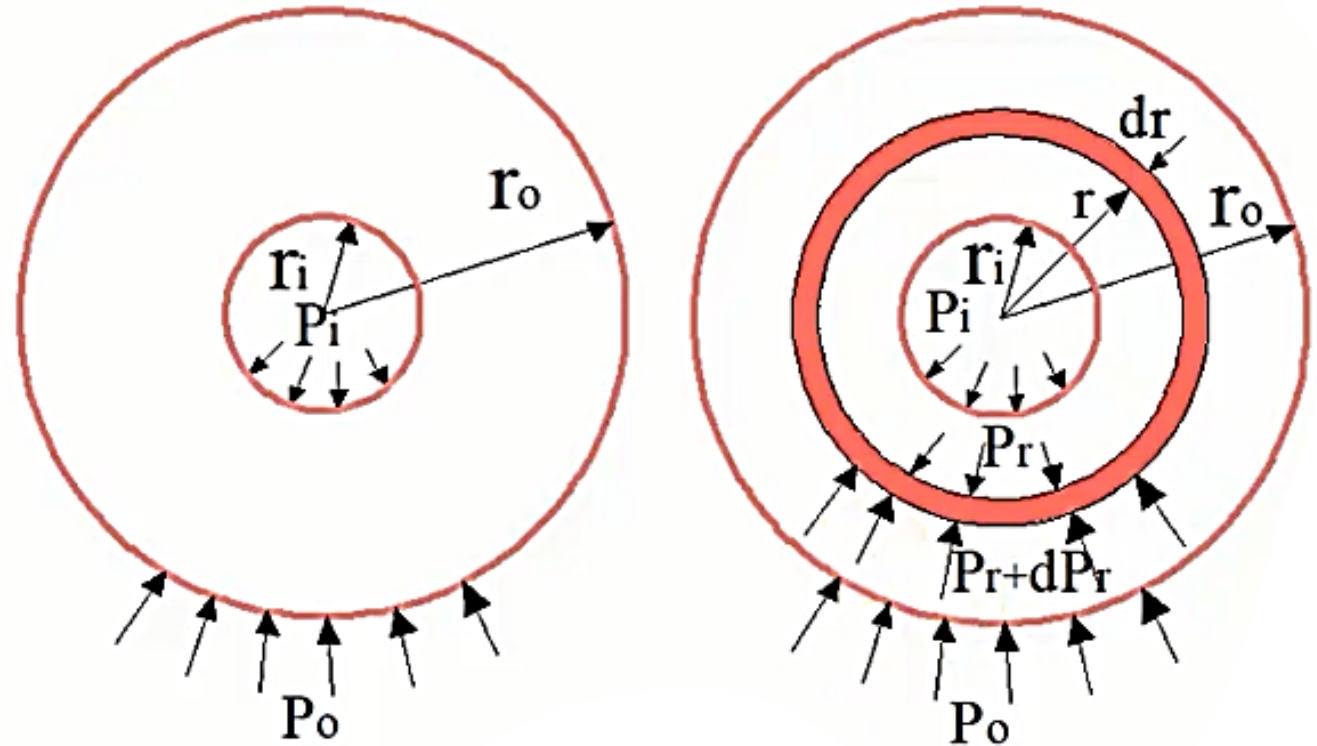
Thick Cylinder



Thick Cylinder

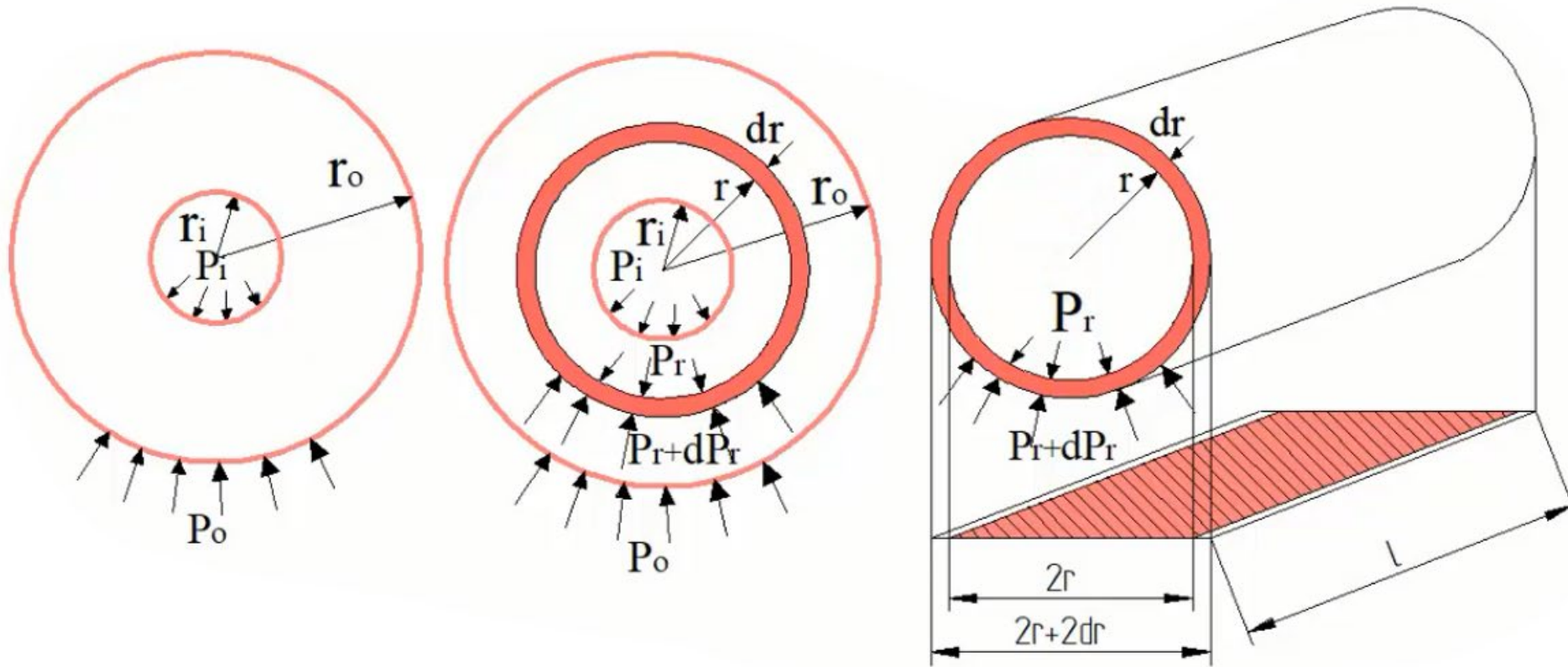
Lame's equation for a thick cylinder

The figure shows a thick cylinder of inside radius r_i and outside radius r_o and length l subjected to internal fluid pressure P_i and external pressure P_o .



Thick Cylinder

Consider a thin cylinder shell at a radius r with a differential thickness dr



Thick Cylinder

The forces due to these pressures are,

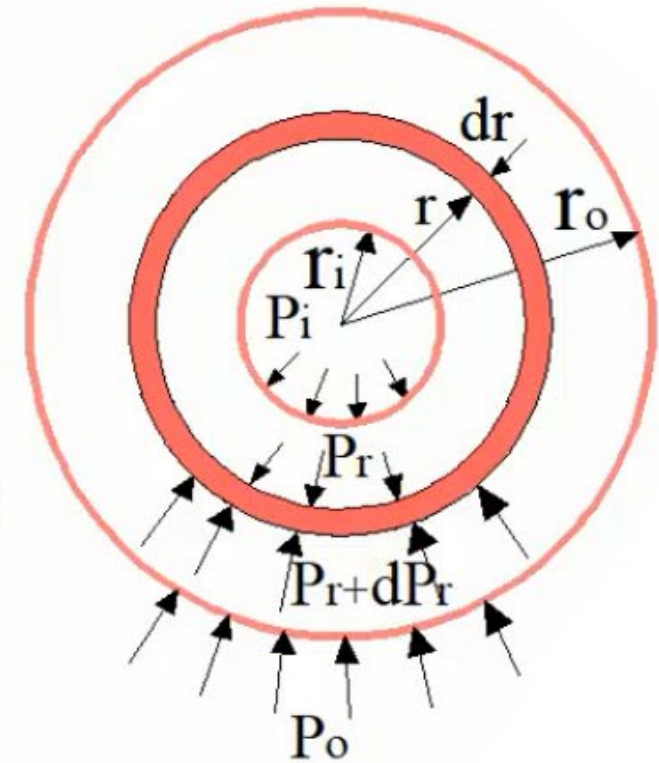
$$\text{From inside } F_i = P_r(2rl)$$

$$\text{From inside } F_o = (P_r + dp_r)(2r + 2dr)l$$

$$\begin{aligned}\text{Bursting force} &= P_r(2rl) - (P_r + dp_r)(2r + 2dr)l \\ &= P_r 2rl - P_r 2rl - P_r 2drl - dp_r 2rl - dp_r 2drl \\ &= -P_r 2drl - dp_r 2rl\end{aligned}$$

$$\sigma_c 2drl = -P_r 2drl - dp_r 2rl$$

$$\text{Hoop stress, } \sigma_c = \frac{-P_r 2drl - dp_r 2rl}{2drl}$$



Thick Cylinder

$$\sigma_c = -P_r - r \frac{dp_r}{dr} \text{ ----- (1)}$$

Longitudinal strain induced in the cylinder is given by

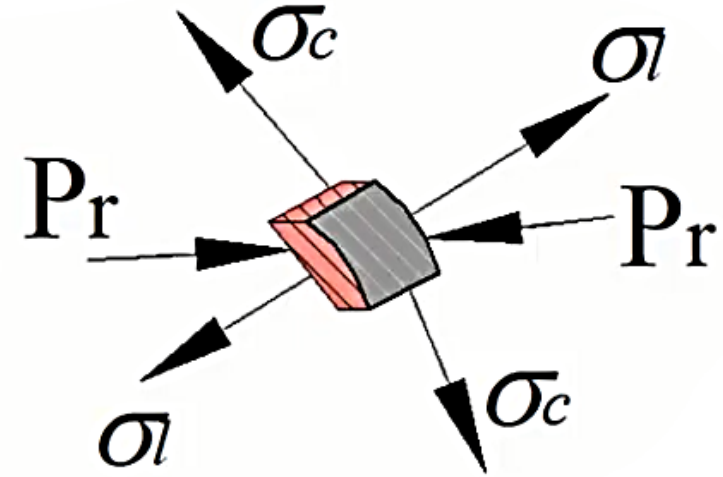
$$e_l = \frac{\sigma_l}{E} - \nu \frac{\sigma_c}{E} + \nu \frac{P_r}{E}$$

The stress induced in the longitudinal direction(σ_l) and longitudinal strain(e_l) are almost constant

$$\therefore -\nu \frac{\sigma_c}{E} + \nu \frac{P_r}{E} = \text{constant}$$

$$\sigma_c - P_r = \text{constant} = 2a$$

$$\sigma_c = P_r + 2a \text{ ----- (2)}$$



Thick Cylinder

Substituting the value of σ_c in equation (1)

$$P_r + 2a = -P_r - r \frac{dp_r}{dr}$$

$$2P_r + 2a = -r \frac{dp_r}{dr}$$

$$2(P_r + a) = -r \frac{dp_r}{dr}$$

$$\frac{dp_r}{(P_r + a)} = -2 \frac{dr}{r}$$

Integrating

$$\log_e (P_r + a) = -2 \log_e r + \text{constant}$$

$$\log_e (P_r + a) = -2 \log_e r + \log_e b \quad \text{constant is written as } \log_e b$$

Thick Cylinder

These equations are known as Lamé's Equation for thick cylinder.

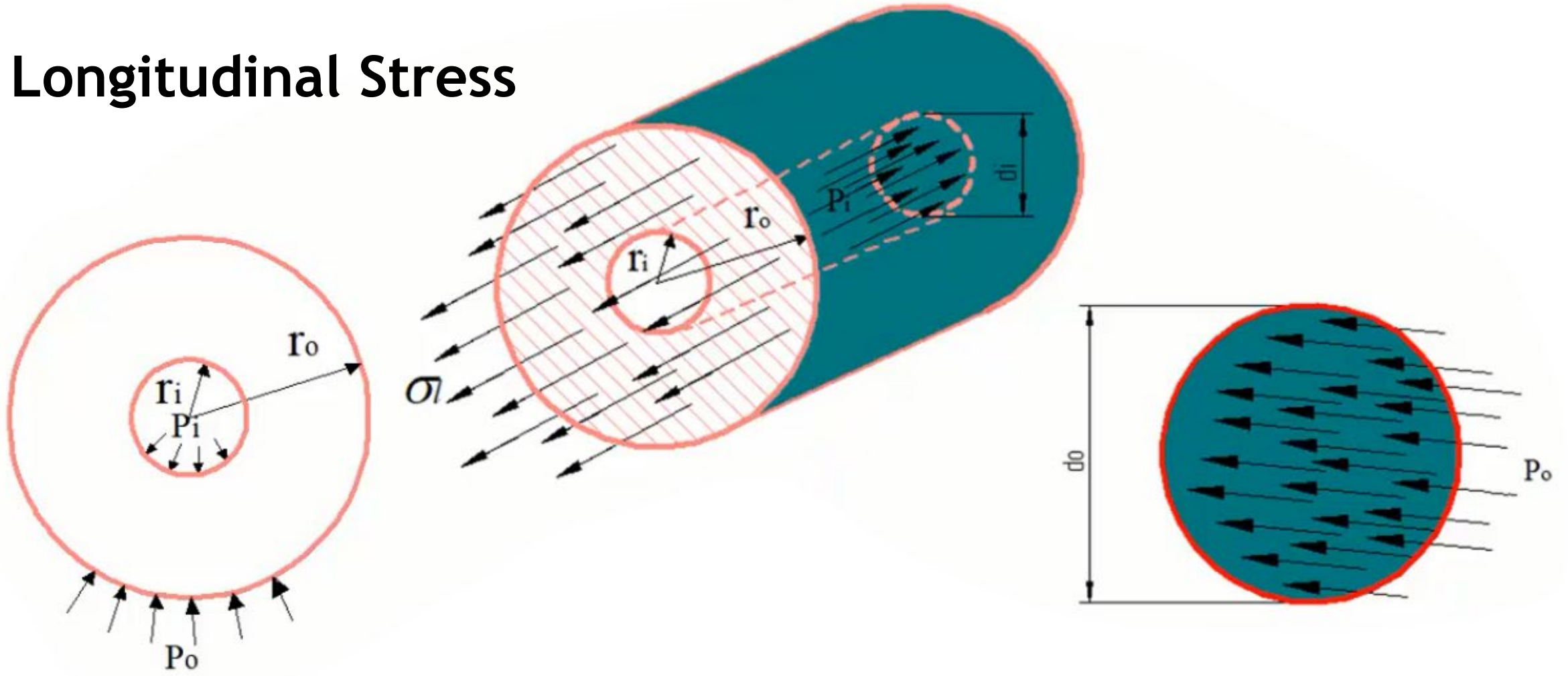
Lamé's Equations for thick cylinder

$$P_r = \frac{b}{r^2} - a$$

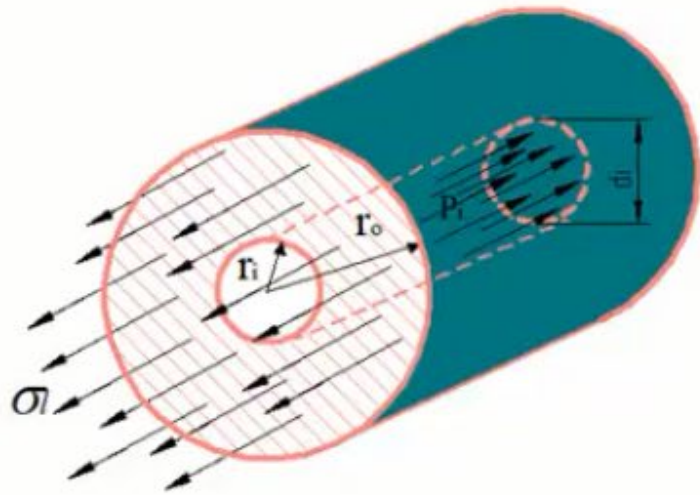
$$\sigma_c = \frac{b}{r^2} + a$$

Thick Cylinder

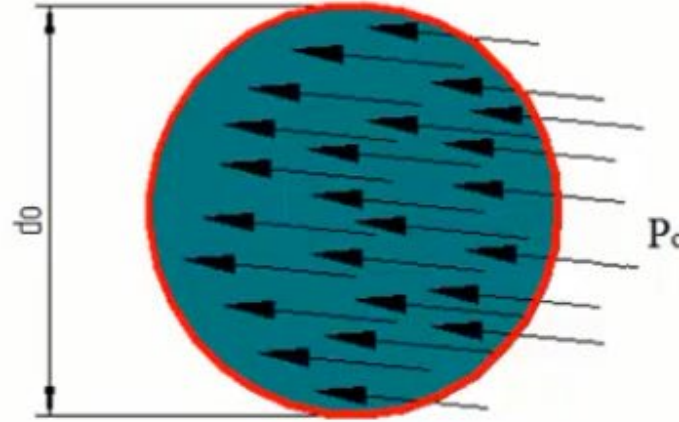
Longitudinal Stress



Thick Cylinder



$$F_i = P_i \times \frac{\pi}{4} d_i^2$$



$$F_o = P_o \times \frac{\pi}{4} d_o^2$$

$$F_l = \left(P_i \times \frac{\pi}{4} d_i^2 - P_o \times \frac{\pi}{4} d_o^2 \right)$$

Resisting area, $A = \frac{\pi}{4} (d_o^2 - d_i^2)$

Thick Cylinder

Longitudinal stress, $\sigma_l = \frac{F_l}{A}$

$$\sigma_l = \frac{\left(P_i \times \frac{\pi}{4} d_i^2 - P_o \times \frac{\pi}{4} d_o^2 \right)}{\frac{\pi}{4} (d_o^2 - d_i^2)}$$

$$\sigma_l = \frac{(P_i d_i^2 - P_o d_o^2)}{(d_o^2 - d_i^2)} = \frac{(P_i r_i^2 - P_o r_o^2)}{(r_o^2 - r_i^2)}$$

Thick Cylinder

For cylinder subjected to only internal pressure, $P_o=0$

$$\sigma_l = \frac{P_i d_i^2}{(d_o^2 - d_i^2)} = \frac{P_i r_i^2}{(r_o^2 - r_i^2)}$$

Circumferential and longitudinal strain

$$e_c = e_d = \frac{\sigma_c}{E} - \nu \frac{\sigma_l}{E} + \nu \frac{P_r}{E} \text{ (vary with radius)}$$

$$e_l = \frac{\sigma_l}{E} - \nu \frac{\sigma_c}{E} + \nu \frac{P_r}{E} = \text{constant}$$

Example 5

A thick cylinder with internal diameter 200mm and 50mm thick wall is subjected to an internal fluid pressure of 50MPa. Find the maximum Hoop stress and plot the variation of Hoop stress and radial pressure across the cylinder wall. Also find the longitudinal stress induced on the cylinder wall.

Inside radius $r_i = 200/2 = 100\text{mm}$

Outside radius $r_o = \text{Inside radius} + \text{thickness} = 100 + 50 = 150\text{mm}$

Inside pressure $P_i = 50 \text{ MPa}$

Example 5

Outside pressure $P_o=0$ (Since it is not mentioned and without having at least one additional data problem cannot be solved)

We know for a thick cylinder, $P_r = \frac{b}{r^2} - a$

Inside pressure $P_i=50$ MPa at radius $r_i=100$ mm

$$50 = \frac{b}{100^2} - a \quad \text{-----(1)}$$

Outside pressure $P_o=0$ at radius $r_o=150$ mm

$$0 = \frac{b}{150^2} - a \quad \text{-----(2)}$$

Subtracting equation (2) from equation (1)

$$50 = \frac{b}{100^2} - \frac{b}{150^2}$$

$$\text{solving } b = 900,000$$

Substituting the value of b in equation (1) or (2)

$$a = 40$$

These values a & b can be used to find the inside Hoop stress and outside Hoop stress.

$$\text{Hoop stress, } \sigma_c = \frac{b}{r^2} + a$$

$$\sigma_c = \frac{900000}{r^2} + 40$$

Inside $r = r_i = 100\text{mm}$

Inside Hoop stress (Max Hoop stress),

$$\sigma_c = \frac{900000}{100^2} + 40 = 130\text{MPa}$$

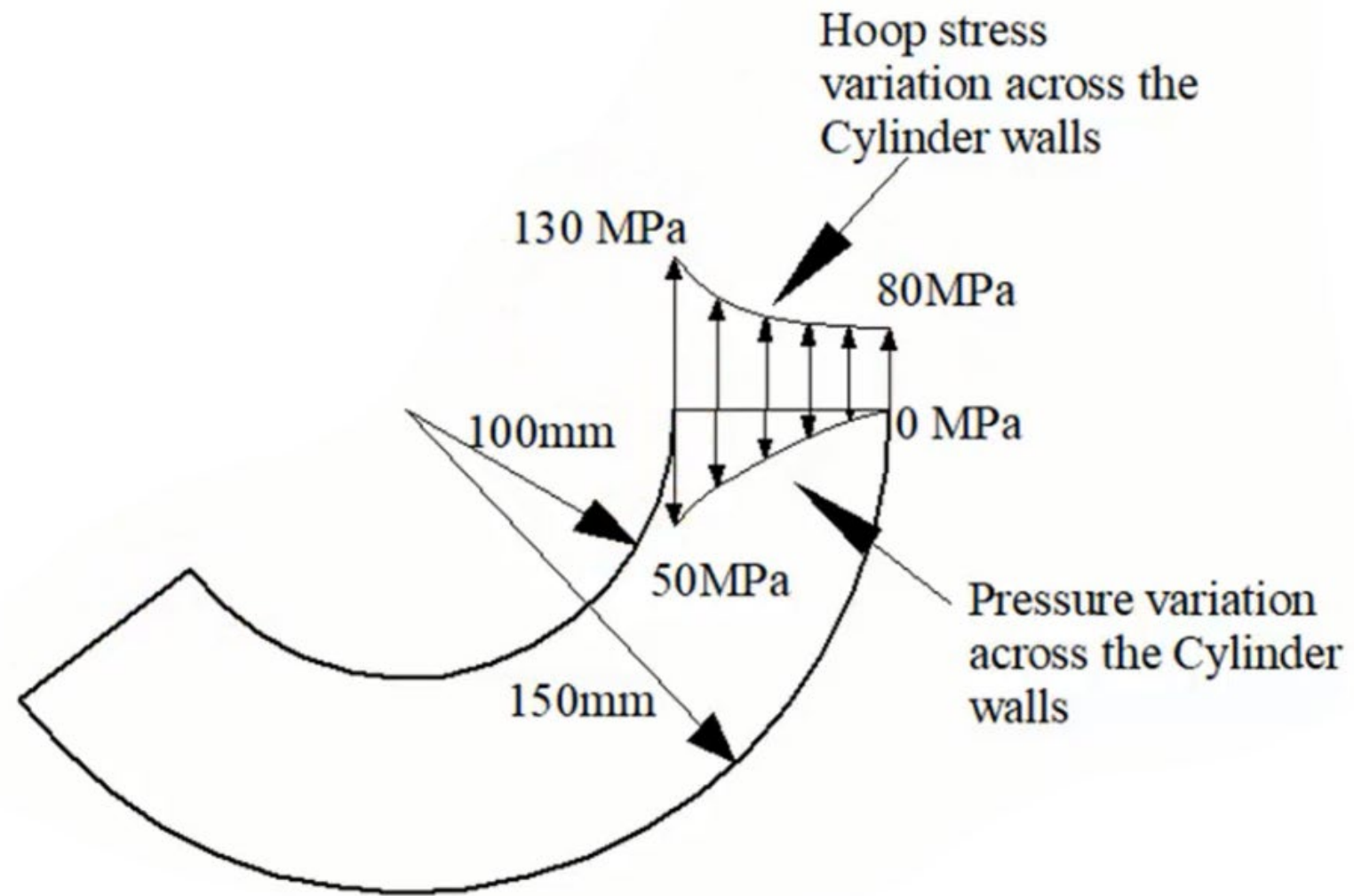
Outside $r = r_o = 150\text{mm}$

Outside Hoop stress,

$$\sigma_c = \frac{900000}{150^2} + 40 = 80\text{MPa}$$

Longitudinal stress induced on the cylinder wall

$$\begin{aligned}\sigma_l &= \frac{(P_i r_i^2 - P_o r_o^2)}{(r_o^2 - r_i^2)} \\ &= \frac{(50 \times 100^2 - 0)}{(150^2 - 100^2)} \\ &= 40 \text{ MPa}\end{aligned}$$



Problem

A thick cylinder 340 mm internal diameter and 600 mm external diameter has to withstand an internal fluid pressure of 65MPa. Find the minimum external pressure required if the maximum tensile stress induced in the cylinder is limited to 82MPa. Plot the variation of Hoop stress and pressure across the cylinder walls.

Inside radius $r_i = 340/2 = 170\text{mm}$

Outside radius $r_o = 600/2 = 300\text{mm}$

Inside pressure $P_i = 50 \text{ Mpa}$

Maximum Hoop stress $\sigma_{\max} = 82\text{MPa}$

Maximum tensile stress is Hoop stress and its maximum always occurs on the inside walls of the cylinder.

Outside pressure $P_o = ?$

We know for a thick cylinder, $P_r = \frac{b}{r^2} - a$ & *Hoop stress*, $\sigma_r = \frac{b}{r^2} + a$

Inside pressure $P_i = 65$ MPa and the corresponding radius, $r_i = 170$ mm

Substituting this in the above equation for radial pressure,

$$65 = \frac{b}{170^2} - a \quad \dots(i)$$

Inside Hoop stress $\sigma_i = 82\text{MPa}$ and the corresponding radius $r_i = 170\text{mm}$

Substituting this in the equation of Hoop stress we get,

$$82 = \frac{b}{170^2} + a \quad \dots(ii)$$

Adding equation (i) and (ii)

$$147 = \frac{2b}{170^2}$$

solving $b = 2124150$

Substituting the value of b in equation (i) or (ii)

we get, $a = 8.5$

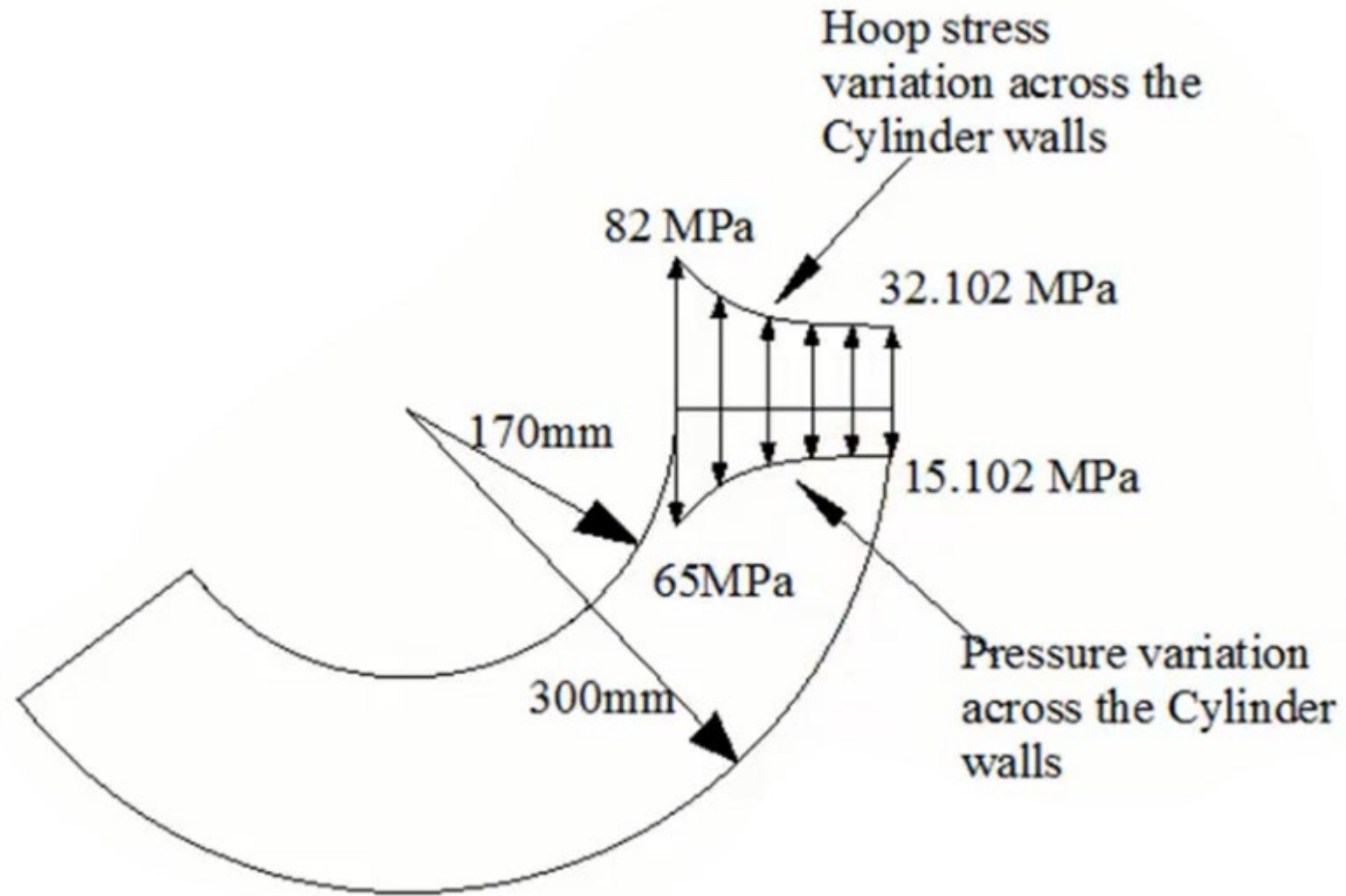
These values a & b can be used to find the external Hoop stress and outside pressure required.

$$\text{Outside Hoop stress, } \sigma_c = \frac{b}{r^2} + a$$

$$\text{Outside } r = r_o = 300\text{mm}$$

$$\sigma_c = \frac{2124150}{300^2} + 8.5 = 32.102\text{MPa}$$

$$\text{Outside Pressure, } P_o = \frac{2124150}{300^2} - 8.5 = 15.102\text{MPa}$$



Problem

A thick cylinder of internal diameter 160mm is subjected to an internal pressure of 40N/mm^2 . If the allowable stress in the material is 120 N/mm^2 , find the required wall thickness of the cylinder.

Inside radius $r_i = 160/2 = 80\text{mm}$

Inside pressure $P_i = 40\text{N/mm}^2$

Allowable stress = 120 N/mm^2

(Since the maximum stress occurs on the inner walls)

Outside pressure $P_o = 0$

We know for a thick cylinder, $P_r = \frac{b}{r^2} - a$

and Hoop stress, $\sigma_c = \frac{b}{r^2} + a$

Inside pressure $P_i = 40 \text{ N/mm}^2$ at radius $r_i = 80 \text{ mm}$

Substituting this in the above equation for radial pressure,

$$40 = \frac{b}{80^2} - a \quad \dots(i)$$

Inside Hoop stress $\sigma_c = 120 \text{ N/mm}^2$ at radius $r_i = 80 \text{ mm}$

Substituting this in the equation for Hoop stress,

$$120 = \frac{b}{80^2} + a \quad \dots(ii)$$

Now adding equation 1 & 2

$$160 = \frac{2b}{80^2}$$

$$\therefore b = 512000$$

Substituting the value of b in equation (i) or (ii)

$$a = 40$$

Outside pressure $P_o=0$ and the corresponding radius r_o to be found
Substituting this in the equation of radial pressure we get,

$$0 = \frac{b}{r_o^2} - a$$

$$0 = \frac{512000}{r_o^2} - 40$$

$$r_o = 113.14mm$$

$$\begin{aligned}\text{Thickness} &= r_o - r_i \\ &= 113.14 - 80 \\ &= 33.14mm\end{aligned}$$

Problem

A cylindrical pressure vessel has inner and outer radii of 200mm and 250mm respectively. The material of the cylinder has an allowable normal stress of 75MN/m^2 . Determine the maximum internal pressure that can be applied and draw a sketch of radial pressure and circumferential stress distribution.

Inside radius $r_i = 200\text{mm}$

Outside radius $r_o = 250\text{mm}$

Allowable stress $= 75\text{ MN/m}^2 = 75\text{ N/mm}^2$

Outside pressure $P_o = 0$

Inside Hoop stress $\sigma_c = 75\text{N/mm}^2$ and the corresponding radius $r_i = 200\text{mm}$

Substituting this in the equation for Hoop stress,

$$75 = \frac{b}{200^2} + a \quad \dots(i)$$

Outside pressure $P_o = 0$

$$0 = \frac{b}{250^2} - a \quad \dots(ii)$$

Now adding equation (i) & (ii)

$$75 = \frac{b}{200^2} + \frac{b}{250^2} \quad \text{solving we get, } b = 1829268.29$$

Substituting the value of b in equation (i) or (ii)

$$a = 29.268$$

To determine the maximum internal pressure substitute the values of a and b in the equation for pressure corresponding to radius r_i .

$$P_i = \frac{1829268.29}{200^2} - 29.268 \quad \text{or} \quad \text{use } P_r = \sigma_c - 2a$$

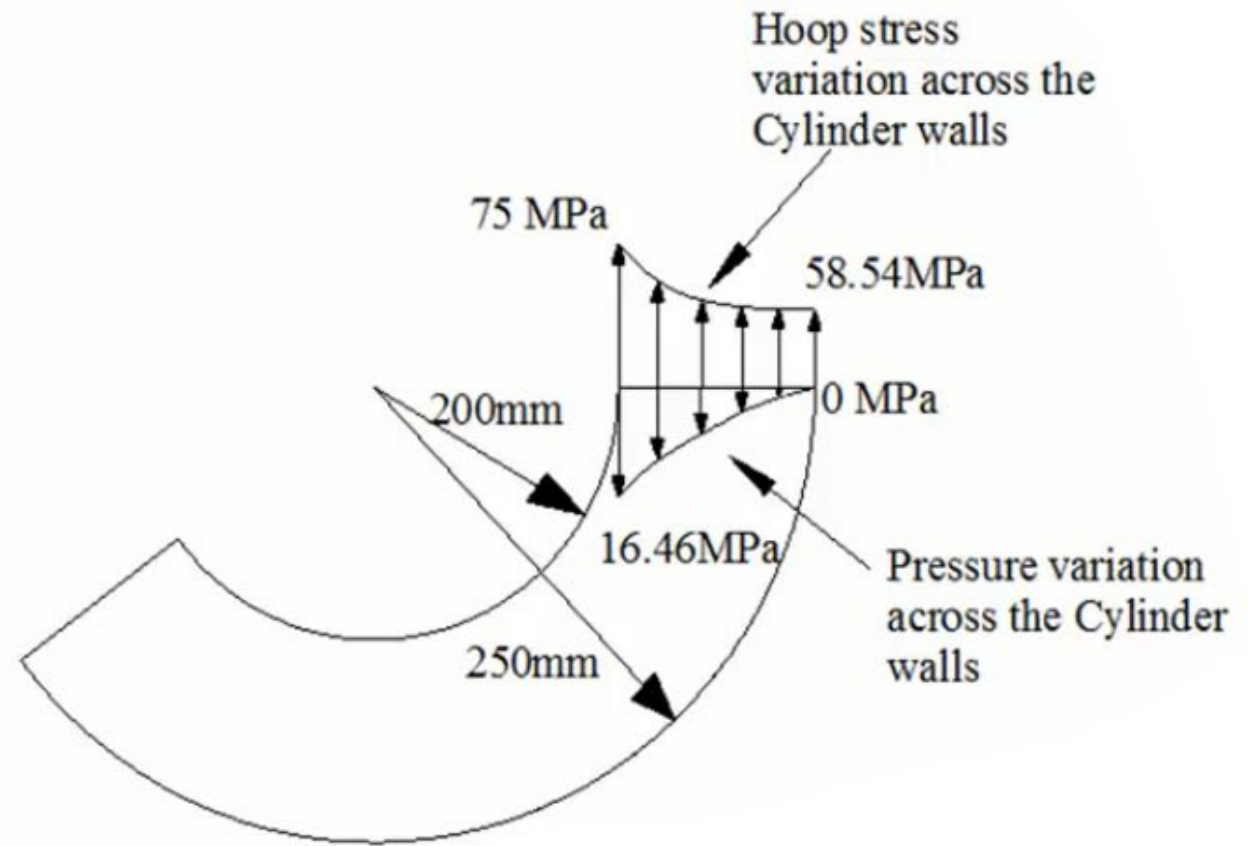
$$P_i = 16.46 \text{ N} / \text{mm}^2$$

Stress on the outer wall of the cylinder

$$\begin{aligned}\sigma_{\min} &= \frac{1829268.29}{250^2} + 29.268 \\ &= 58.54 \text{ N / mm}^2\end{aligned}$$

Or we may use

$$\begin{aligned}\sigma_{\min} &= P_r + 2a \\ &= 0 + 2 \times 29.268 \\ &= 58.54 \text{ N / mm}^2\end{aligned}$$





Thank you

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